

Sample-Efficient Seed Selection under Overexposure: Estimating Marginal Gains without a Coverage Representation

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Abstract. Influence maximization is dominated by reverse-reachability (RR) sampling, whose guarantees rest on a coverage representation: for a random set R independent of the seed set S , the spread is $c \cdot \Pr[S \cap R \neq \emptyset]$, a non-decreasing submodular function of S . We study targeted influence maximization under a threshold-dependent overexposure model in which a node becomes positively active only when the weighted influence it receives from its in-neighbours lands inside a node-specific window $[\theta_v^*, \theta_v^+]$. We first delimit what is and is not ruled out here. On a one-hop instance with two candidates of weight 0.4 into a single target, the model’s own window distribution gives $F_D(\{a\}) = 0.48$ and $F_D(\{a, b\}) = 0.32$: the objective is non-monotone, so no non-negative seed-independent coverage function can match it, and the standard RR representation is unavailable. The statement is deliberately narrow — it does not exclude signed decompositions, restricted-instance algorithms, or Monte-Carlo estimation, and it does not imply that learning must help. It also requires naming the threshold distribution: under uniform-threshold linear thresholds the same instance is monotone, so the two readings cannot be conflated. We then report the regime where this matters, re-measured under the corrected activation rule at 300 paired Monte-Carlo trials per cell: saturation is controlled by the seed *fraction* $|S|/n$ rather than the absolute budget, and across eight graphs spanning mean degree 4.2 to 56.0, out-degree is negatively correlated with the true marginal gain on seven of them once $|S|/n \geq 20\%$, while a training-free state-conditioned closed form is the better ranker on all eight. We also report the corrected negative-share measurement: the share of candidates with a negative marginal gain is 0.5% on average and at most 4.0% in any saturated cell at $MC = 300$; the low-budget predecessor is not used as evidence. Finally we give a measurement protocol for the model — five checks covering state transitions, threshold-law identification, Monte-Carlo noise, state-size pooling, and candidate-pool repetition. We report both what the evidence supports and what it does not.

Keywords: Influence maximization · Overexposure · Marginal gain estimation · Simulation budget · Negative results

1 Introduction

Influence maximization (IM) asks for a seed set of size k that maximises expected influence spread. The dominant algorithmic family is reverse-reachability (RR) sampling [1, 11, 9]: sample random reverse walks, observe that $\mathbb{E}[\sigma(S)] = n \cdot \Pr[S \cap \mathcal{R} \neq \emptyset]$, and solve a maximum-coverage problem on the sampled sets. RR is attractive because it is both fast in practice and provably near-optimal. Its guarantees, however, rest on a specific generative structure: an independent per-edge random mechanism (the live-edge graph of independent cascade, or the trigger set of linear thresholds) together with a monotone submodular spread function.

This paper is about a setting where that structure is absent, and about what a practitioner should do instead. We study *targeted* IM under a threshold-dependent overexposure model [13] motivated by advertising fatigue: a user becomes positively active only when the weighted influence they receive from their in-neighbours lands inside a personal window $[\theta_v^\kappa, \theta_v^\tau]$. Too little influence leaves them inactive; too much makes them *negatively* active, and a negatively active user never promotes the item. The window is drawn uniformly from the simplex $\{0 \leq \theta^\kappa \leq \theta^\tau \leq 1\}$, and the exposure a node accumulates is the sum of the weights of the in-neighbours that were *ever* positively activated.

Our first contribution delimits what this objective rules out, and what it does not. The coverage representation behind RR sampling requires the spread to be $c \cdot \Pr[S \cap R \neq \emptyset]$ for some random R independent of S ; such a function is non-decreasing and submodular. On a one-hop instance with two candidates of weight 0.4 into a single target the model's own window distribution gives $F_D(\{a\}) = 0.48$ and $F_D(\{a, b\}) = 0.32$, so the target is non-monotone and no such coverage function can match it (§4). We state the scope precisely because it is easy to over-read: this excludes the standard coverage form, and nothing else — not signed decompositions, not algorithms for restricted instances, and not Monte-Carlo estimation, since a sample mean is itself unbiased. It is also a statement about a named threshold distribution: under uniform-threshold linear thresholds the same instance is monotone, so the two readings cannot be conflated. A weaker version of this argument, phrased as "RR sets are empty", appeared in an earlier draft of this work and is withdrawn; the probe it rested on was mis-specified.

Our second contribution is to identify *when* this matters, re-measured under the corrected activation rule at 300 paired Monte-Carlo trials per cell (§6). A naive experiment measures out-degree against the true marginal gain and concludes that degree is fine — we made exactly that mistake, twice. The first time, we swept an absolute budget on a graph where that budget was at most 26% coverage and mostly far less; measured as a *fraction* $|S|/n$, the same sweep tells a different story, so any statement about overexposure that does not report $|S|/n$ should be distrusted. The second time, we read the regime signal out of twelve Monte-Carlo trials per cell. Across eight graphs spanning mean degree 4.2 to 56.0, out-degree is negatively correlated with the true marginal gain on seven of them once $|S|/n \geq 20\%$ — NetHEPT, the sparsest, is the exception — and a training-free two-hop closed form is the better ranker on all eight. But the same

twelve-trial measurement also told us that 8–35% of candidates have a *negative* marginal gain, and that is false: at 300 trials the figure is 0.5% on average with a worst cell of 4.0%. The sign flip is real and the negative-marginal share was Monte-Carlo artefact, and only adequate power separates the two.

An earlier version of this work reported a baseline comparison built on an activation rule that denied an over-exposed node the ability to turn negative and on Monte-Carlo budgets below the threshold required for saturated rank measurements. The corrected rule removes 60–63% of the measured spread on Congress-Twitter, which is larger than every effect reported in that version. We therefore omit those tables from the submission and retain only the protocol lesson: results from this model must be regenerated under the corrected state machine and measurement contract.

Contributions.

- We show that reverse-reachability sampling is not expressible by a coverage representation under threshold-dependent overexposure, by exact arithmetic on a one-hop instance (§4). The statement is deliberately narrow: it excludes the standard non-negative seed-independent coverage form, and nothing else — not signed decompositions, not algorithms for restricted instances, not Monte-Carlo estimation, and not the possibility that learning helps. It also fails under uniform-threshold linear thresholds, which is why the threshold distribution must be named.
- We identify the seed *fraction* as the controlling variable and report a re-measured regime characterisation: across eight graphs, out-degree becomes an anti-signal on seven once $|S|/n \geq 20\%$ while a training-free state-conditioned closed form is the better ranker on all eight (§6). We also withdraw the negative-marginal share we previously reported from twelve trials per cell; at 300 trials it is 0.5% on average and 4.0% at worst.
- We give a measurement protocol for this model (§5) in the form of five checks covering state transitions, threshold-law identification, Monte-Carlo noise, state-size pooling, and candidate-pool repetition. These checks define the evidence contract for the corrected measurements; withdrawn pre-fix magnitudes are not used.
- We show that the upper bound the source model optimises is a *difference* of two submodular functions, so even a valid bound would not by itself license a sampling-based guarantee (§9).
- We state eight protocol confounds we found in our own experiment code — inconsistent stopping under a budget of *at most* k , unpriced state acquisition, a noisy estimator called an oracle, mixed seed sizes in one statistic, a single random candidate pool, splits that do not deduplicate, unreported label uncertainty, and weights normalised beyond what the model allows — and for each we say whether it was reproduced or only alleged (§9).
- We report the corrected state machine, the counterexamples that exposed it, and the measurement confounds that must be controlled before a sequential quality–cost claim can be made (§9).

2 Related Work

Scalable influence maximization. The reverse-reachability family underpins the fastest known IM algorithms. TIM and TIM+ [1, 12] and IMM [11] sample reverse walks and solve maximum coverage, achieving $(1 - 1/e - \varepsilon)$ guarantees with near-linear expected running time; OPIM [10] extends this to the online setting; [9] tightens the stopping rule. GPU-parallel implementations [5, 8] report speedups of two to three orders of magnitude over sequential baselines. All of these inherit the identity $\mathbb{E}[\sigma(S)] = n \cdot \Pr[S \cap \mathcal{R} \neq \emptyset]$, and therefore all of them require a coverage representation that §4 shows cannot match this objective here. Our result is not that these algorithms are slow, but that they cannot be instantiated.

Learning-based influence maximization. A complementary line replaces the influence oracle with a learned predictor, either to prune the candidate set before exact greedy [3] or to estimate marginal gains directly. A recent benchmark study [6] finds that, across a range of IM and maximum-coverage tasks, classical lazy greedy and sampling-based methods remain competitive with or superior to deep reinforcement learning approaches. Our §9 reaches a consistent conclusion in the overexposure setting, via a different route: the training-free closed form is not beaten by the learned scorers we tested.

Overexposure and negative influence. That excessive exposure can be counter-productive is well documented. The model we study is due to [13], who introduce the threshold window and derive the $2\delta(1 - \delta)$ activation probability; they establish non-monotonicity and non-submodularity of the objective and propose an incremental greedy algorithm with a γ/k ratio plus an upper-bound method built on two monotone submodular linear-threshold surrogates. An earlier line models overexposure through a logistic opinion-formation term inside a targeted-IM objective [7], and notes explicitly that because the spread function is non-submodular, the greedy step cannot be accelerated with lazy evaluation or reverse-influence sets. Our contribution relative to this line is to give a structural reason — rather than an empirical observation — for why reverse-influence machinery does not apply, and to characterise the seed-fraction regime in which the resulting failure of degree-based heuristics becomes visible.

Threshold and complex-contagion models. IM has also been studied under threshold dynamics in which adoption requires reinforcement from multiple neighbours. Reinforcement learning has been applied to such settings [2]. Our finding adds a boundary condition to this literature: as soon as the activation rule depends on an accumulated *sum* that must land inside an interval, the per-edge random structure that sampling-based methods rely on is lost, and the guarantees that accompany them do not transfer.

CELF, and the cost of a violated precondition. CELF [4] is the standard lazy-evaluation accelerator for greedy maximisation and is cited as a baseline by

the source model. Its saving comes from a stale-upper-bound argument: if the recomputed gain of the heap head still dominates the runner-up’s stale gain, the head is the true argmax and the remaining candidates can be skipped. That argument requires the objective to be monotone and submodular. The objective here is neither — the source model’s Theorem 2 establishes non-monotonicity and its Theorem 3 establishes that neither submodularity nor supermodularity holds — so a candidate whose gain has fallen can rise again later and the skip test is unsound. We therefore implement CELF as the *exact* greedy it is designed to accelerate; a lazy variant would yield an artificially weaker baseline whose shortfall came from a violated precondition rather than from the algorithm. We raise the point because it generalises: every overexposure objective inherits the same obstruction, and lazy evaluation is one of the few standard accelerations that is *not* blocked by the absence of reverse reachability, so it is worth knowing that it fails for a different reason.

3 Preliminaries

3.1 Targeted influence maximization

Let $G = (V, E)$ be a directed graph with $|V| = n$, and let $\omega_{uv} \in [0, 1]$ be the influence weight of edge (u, v) . A *target set* $T \subseteq V$ is given, and the goal is to choose a seed set $S \subseteq V$ with $|S| \leq k$ maximising the expected number of positively activated nodes in T . Write $\sigma(S)$ for that expectation.

3.2 Threshold-dependent overexposure

Each node v carries a *threshold window* $[\theta_v^\kappa, \theta_v^\tau]$. The pair is drawn uniformly from the $2D$ simplex

$$\Delta = \{(\theta^\kappa, \theta^\tau) \in [0, 1]^2 : \theta^\kappa \leq \theta^\tau\}, \quad (1)$$

that is, with density 2 on Δ . Here θ_v^κ is an activation threshold and θ_v^τ an overexposure threshold.

Write $A_t^{\text{in}}(v)$ for the set of in-neighbours of v that were positively activated at some time $\leq t$. Crucially, this set *keeps* nodes that were later turned negative — their influence is assumed to persist. The *exposure score* of v at time t is

$$\delta(v, t) = \sum_{u \in A_t^{\text{in}}(v)} \omega_{uv}. \quad (2)$$

States are {inactive, positive, negative}. At $t = 0$ the seeds are positive and remain so. At $t + 1$ the exposure of the neighbours of the nodes that turned positive at t is updated, and each such node transitions by

$$v \text{ becomes } \begin{cases} \text{positive} & \text{if } \theta_v^\kappa \leq \delta(v, t+1) \leq \theta_v^\tau, \\ \text{negative} & \text{if } \delta(v, t+1) > \theta_v^\tau, \\ \text{inactive} & \text{otherwise (re-evaluated when more influence arrives).} \end{cases} \quad (3)$$

Transitions are one-way: inactive $\rightarrow$ positive $\rightarrow$ negative, never backwards. A negatively active node does not promote. We call this the *threshold-window* process.

Remark 1 (Activation probability). Integrating over the window draw gives the marginal positive-activation probability as a function of exposure alone,

$$\Pr[\theta^\kappa \leq \delta \leq \theta^\tau] = 2\delta(1-\delta), \quad (4)$$

which we denote $g(\delta)$. This is the quantity the source model builds its analysis on. It is 0 at $\delta \in \{0, 1\}$ and peaks at $\delta = 1/2$.

Two structural facts drive everything that follows and are worth isolating.

Observation 1 (Exposure is monotone in time, the probability is not). *$A_t^{\text{in}}(v)$ grows with t and never shrinks, because a node that has been positively activated keeps contributing even after turning negative. Hence $\delta(v, t)$ is non-decreasing in t for every v . The activation probability $g(\delta) = 2\delta(1-\delta)$ is therefore non-monotone in time: it rises while $\delta < 1/2$ and falls once $\delta > 1/2$. The only source of non-monotonicity in the model is the shape of g , not the exposure accumulation.*

Observation 2 (Exposure is a set function of the seed set). *Condition on a fixed draw of all windows. Then for each v the indicator of “ v was ever positively activated” is a deterministic function of S , and consequently so is $\delta(v, t)$ for every v and t , and so is the final spread. The only randomness in the entire process is the single window draw.*

Observation 2 is the reason the usual sampling machinery has nothing to work with, and we make that precise next.

3.3 Reverse-reachability sampling, and what it requires

RR sampling relies on the identity

$$\mathbb{E}[\sigma(S)] = n \cdot \Pr[S \cap \mathcal{R} \neq \emptyset], \quad (5)$$

where $\mathcal{R}$ is a random set generated by a reverse walk from a uniformly random node. The identity is available whenever the process admits a *randomised graph structure* over edges — the live-edge graph in independent cascade, the trigger set in linear thresholds — such that v is active iff it is reachable from S in that structure. Two features are required: (i) reachability can start from a base case in which a node is active under *incoming influence from a single edge or a single predecessor*, and (ii) the structure is sampled independently of S .

4 What the Objective Rules Out, and What It Does Not

An earlier version of this work argued that reverse-reachability sampling has no domain here because every RR set is empty. That argument is withdrawn. The probe it rested on required a predecessor’s window to contain zero and excluded the root, and it tested whether the returned set was non-empty rather than whether it intersected the seed set. A reverse-reachability set for a root answers *which seeds can reach this root*; it does not require a node to self-activate under zero incoming influence. The empty sets were an artefact of a mis-specified probe.

What survives is a strictly narrower statement, proved by exact arithmetic on a one-hop instance rather than by simulation.

4.1 The instance

Take a single target $D = \{t\}$ and two candidates a, b with edges $a \rightarrow t$ and $b \rightarrow t$, both of weight $\omega = 0.40$, and seeds drawn from $V \setminus D$.

Under the model’s own window distribution – $(\theta^\kappa, \theta^\tau)$ uniform on the 2-simplex $\{0 \leq \theta^\kappa \leq \theta^\tau \leq 1\}$ – the target’s positive-activation probability is $g(\delta) = 2\delta(1 - \delta)$ with δ the exposure it receives, so

S	δ	$F_D(S)$ under the model	$F_D(S)$ under uniform-threshold LT
$\emptyset$	0.00	0.0000	0.0000
$\{a\}$	0.40	0.4800	0.4000
$\{a, b\}$	0.80	0.3200	0.8000

Two things follow, and they must be kept apart.

The objective is non-monotone. $F_D(\{a, b\}) = 0.3200 < F_D(\{a\}) = 0.4800$. Adding a seed can strictly decrease the target’s activation probability.

No non-negative seed-independent coverage function matches it. For any random set R independent of S , the function $S \mapsto c \cdot \Pr[S \cap R \neq \emptyset]$ is non-decreasing and submodular in S , because adding a seed can only create intersections and the marginal value of a seed cannot increase as S grows. A non-monotone F_D therefore cannot equal such a function on this instance, for any c and any distribution of R .

4.2 Scope

The statement above excludes exactly one thing: the standard, non-negative, seed-independent coverage representation that underlies RR sampling. It does not exclude, and should not be read as excluding:

- signed decompositions, or sampling structures other than non-negative coverage;
- algorithms specialised to restricted instance classes;
- Monte-Carlo estimation. A plain sample mean is an unbiased estimator of F_D and is the reference used throughout this paper;
- any claim that a learned predictor must help. Whether learning adds anything is an empirical question this paper treats as open.

4.3 Why the threshold distribution must be stated explicitly

The table also shows why the two threshold readings cannot be conflated. Under uniform-threshold LT with $\theta^\kappa \sim U[0, 1]$ and no upper threshold, the same instance is *monotone* ($0.4000 \rightarrow 0.8000$), so the argument would fail there. The source model samples $(\theta^\kappa, \theta^\tau)$ jointly from the simplex, which induces the marginal $F^\kappa(x) = 2x - x^2$ for the lower threshold and $F^\tau(x) = x^2$ for the upper; a later lemma in the same work describes sampling θ^τ uniformly and then θ^κ conditionally, which is a different joint law. Every claim in this paper therefore names the distribution it assumes, and the degenerate no-overexposure setting is treated as its own configuration rather than as standard uniform-threshold LT.

The practical consequence for the rest of the paper is that this objective must be estimated by simulating the process. Sampling-based coverage machinery is unavailable, so the question turns into how to spend a simulation budget well – which is the subject of §8.

5 A Measurement Protocol for This Model

Everything in this section was found by getting it wrong first. Each item states the trap, the failure mode it can produce, and the check that catches it. Historical pre-fix magnitudes are not part of the submission artifact; the corrected protocol is the fact source for any future measurement.

5.1 Re-evaluate every non-seed node every round

The trap. Recording a node as “settled” the first time it activates. Under this model a node’s state is a function of the exposure it has received *so far*, and exposure keeps growing as its in-neighbours activate. A node that is positive in round t can be over-exposed in round $t + 1$, and the model says it then becomes negative.

The error. A counterexample with no Monte-Carlo noise: edges $s \rightarrow a$ with weight 0.4, $s \rightarrow b$ with weight 0.5, $b \rightarrow a$ with weight 0.4; seed $\{s\}$; windows $a = [0.2, 0.6]$ and $b = [0.2, 0.9]$. Node a first receives $\delta = 0.4$, inside its window, so it turns positive. Node b then activates and raises a ’s exposure to $0.9 > 0.6$, so a must turn negative. Freezing a gives spread 3 and an empty negative set; the correct answer is spread 2 with a negative.

Why it matters more than it looks. On the audit graph the freezing error changes the measured spread by far more than the policy differences that the paper is intended to compare. Those pre-fix numerical measurements are omitted from the submission; the point retained here is the mechanism, not a withdrawn benchmark.

The check. The transition rule must be the literal one, $\theta_v^\kappa \leq \delta(v, t) \leq \theta_v^\tau$, evaluated for every non-seed node in every round. Two boundary cases catch most implementations: a node with $\delta = \theta^\tau = 1$ is *positive* (the comparison is inclusive), and a node whose δ exceeds θ^τ after having been positive must be moved out of the positive set.

5.2 State the threshold law, because the sign of the effect depends on it

The trap. Sampling two independent uniforms and clamping the upper one to 1, then describing the result as “linear thresholds”.

The error. The model draws $(\theta^\kappa, \theta^\tau)$ uniformly from the simplex $\{0 \leq \theta^\kappa \leq \theta^\tau \leq 1\}$, i.e. as the order statistics of two independent uniforms. The induced marginals are

$$F^\kappa(x) = 2x - x^2, \quad F^\tau(x) = x^2,$$

so $\mathbb{E}[\theta^\kappa] = 1/3$ and $\mathbb{E}[\theta^\tau] = 2/3$. Clamping $\theta^\tau = 1$ keeps $F^\kappa = 2x - x^2$, but replacing the joint draw with $\theta^\kappa \sim U[0, 1]$ changes the lower marginal to $\mathbb{E}[\theta^\kappa] = 1/2$. The two are different processes: on a one-hop instance with two candidates of weight 0.4 into a single target, the model’s law gives $F_D(\{a\}) = 0.4800 > F_D(\{a, b\}) = 0.3200$ — non-monotone — while uniform-threshold LT gives $0.4000 < 0.8000$, which is monotone. Every statement about non-monotonicity, and therefore about the inapplicability of coverage sampling, holds for one law and fails for the other.

The check. Name the law in the configuration, and verify the realised marginals against $2x - x^2$ and x^2 rather than trusting the sampler. Four laws must be distinguished, not two: the simplex law; the same draw with θ^τ clamped (overexposure off, κ marginal preserved); genuine uniform-threshold LT; and the simplex law with θ^τ ’s support raised, which is an intervention knob rather than a degeneracy.

5.3 Do not quote a rank correlation below MC = 300 in the saturated regime

The trap. Measuring candidate marginals with a few dozen Monte-Carlo trials and reporting rank correlations, or a share of candidates with a negative marginal.

The error. In the regime this paper is about, the true marginals can be small compared with the error of estimating them. Low-trial labels can therefore change sign and rank ordering as the Monte-Carlo budget increases. The pre-fix numerical sweep is withdrawn; only the measurement lesson is retained here.

The check. Report the paired standard error of every label. A candidate is negative only if its mean is more than two paired standard errors below zero, and a share of negative candidates is only interpretable with that column present. Prefer paired spread comparisons, which cancel the window-draw noise by construction and are unaffected by this trap.

5.4 Never pool seed sizes in a state-dependence statistic

The trap. Measuring whether a candidate’s marginal depends on the current seed set by drawing contexts at several seed sizes — typically $0, k, 2k, 3k$ — and computing one within/between variance ratio over all of them.

The error. The unsaturated context can dominate the pooled statistic. A pooled ratio therefore reports behaviour outside the regime of interest and can give the right qualitative verdict for the wrong reason. We now compute the ratio inside one seed size and report it per size.

The check. Compute the ratio inside one seed size and report it per size. A cell counts as state dependent only if it clears the threshold at the seed size at which the rest of the measurement was taken.

5.5 Repeat the candidate pool, because one draw can flip the sign

The trap. Drawing one random pool of candidates and reporting the headroom a policy has on it.

The error. The pool’s degree profile decides how much headroom is visible, and the *sign* of the policy gap can depend on the draw. A single-pool number therefore cannot support a claim in either direction.

The check. Draw the pool degree-stratified so its profile matches the graph’s by construction, repeat over several independent draws, and report the median with a min–max band. A verdict should not be published from an unfinished sweep: if a run covers a subset of the requested graphs, say so in the result file rather than reporting a partial aggregate.

5.6 What these five checks cost

They are cheap. Re-evaluating non-seed nodes adds one pass over the frontier per round. Naming the threshold law is free. Storing a paired standard error alongside each label costs nothing, because the paired differences are formed before averaging anyway. Computing the state-dependence ratio per size rather than pooled is a loop boundary. Stratifying and repeating the pool multiplies one measurement by the number of draws. The expensive check is the Monte-Carlo budget, which is a factor of 25 over a low-budget run. The earlier low-budget tables are omitted from the submission; the remaining experiment section records the protocol boundary and the pending post-fix quality–cost study.

6 The Controlling Variable Is the Seed Fraction

6.1 Why absolute budgets mislead

Our first two attempts to characterise this problem produced the conclusion that out-degree *does not* fail under overexposure, and both were wrong for the same reason. We sampled candidate nodes uniformly from the whole graph and correlated out-degree with the true marginal gain. On a heavy-tailed graph such a pool consists almost entirely of low-degree nodes, so the correlation is dominated by the global regularity “high-degree nodes are more valuable” rather than by any ability to rank candidates under a given state. Worse, the experiments were run at tiny coverage: on NetHEPT ($n = 15233$) a budget of 200 seeds is 1.3% of the network.

The decisive variable is not the budget but the *seed fraction* $|S|/n$. Saturation is a property of coverage, not of the number of seeds: the same budget of 200 seeds is 1.3% of NetHEPT but 42% of Congress-Twitter ($n = 475$).

6.2 Sweeping the fraction across eight graphs

We sweep $|S|/n \in \{0.10, 0.20, 0.40\}$ on eight graphs whose mean degree spans 4.23 to 55.95, and measure the rank correlation between out-degree and the true marginal gain $\Delta(v \mid S) = \sigma(S \cup \{v\}) - \sigma(S)$. Every cell uses paired Monte Carlo with the threshold windows shared across candidates within a trial, so the *differences* between candidates carry no window-draw noise.

Two protocol decisions from §5 are load-bearing here. The measurement is taken at $MC = 300$ paired trials, because at 60 trials the label noise in the saturated regime is as large as the dispersion being measured. And each candidate’s label carries its own paired standard error, so the “neg.” columns below can separate a candidate that is negative from one that is merely noisy.

Three observations follow, and the third is a correction of an earlier version of this work.

1. **Out-degree flips sign.** It is positively correlated with the marginal gain on only one of the eight graphs — NetHEPT, the sparsest, at $\langle k \rangle = 4.23$ — and negatively on the other seven once $|S|/n \geq 20\%$. Averaged over the saturated cells the correlation is -0.177 for out-degree against $+0.420$ for the closed form δ_2 .
2. **The state-conditioned closed form is the better ranker on all eight.** It is the only scorer we tested that is positive in every saturated cell, and its advantage is largest exactly where out-degree is most misleading.
3. **The corrected negative-share result is small.** Measured at $MC = 300$, the mean share over saturated cells is 0.5%, the worst cell in the entire sweep is 4.0%, and the mean share negative by more than two paired standard errors is 0.2%. The low-budget predecessor is excluded because its label noise is not a valid basis for this claim.

Table 1. Out-degree versus the state-conditioned closed form, re-measured under the corrected activation rule at $MC = 300$ paired trials per cell. Values are means over $|S|/n \geq 20\%$. ρ is the Spearman correlation of the score with the true marginal gain; “neg.” is the share of candidates with a negative marginal and “neg.*” the share negative by more than two paired standard errors. Out-degree is negatively correlated on 7 of 8 graphs and δ_2 is the better ranker on 8 of 8.

graph	n	$\langle k \rangle$	ρ_{degree}	ρ_{δ_2}	neg.	neg.*
NetHEPT	15233	4.23	+0.099	+0.693	1.0%	1.0%
p2p-Gnutella08	6301	6.59	−0.052	+0.400	0.0%	0.0%
ca-GrQc	5242	11.06	−0.583	+0.640	0.0%	0.0%
Wiki-Vote	7115	29.15	−0.042	+0.386	0.0%	0.0%
ca-HepPh	12008	39.48	−0.137	+0.386	0.0%	0.0%
Facebook	4039	43.69	−0.419	+0.527	2.0%	1.0%
email-Eu-core	1005	50.89	−0.236	+0.239	0.0%	0.0%
Congress-Twitter	475	55.95	−0.047	+0.085	1.0%	0.0%

What this means for the argument. Non-monotonicity is *possible* in this model — §4 exhibits a one-hop instance where adding a seed strictly lowers the target’s activation probability, and that instance is exact arithmetic, not simulation. But it is rare on the graphs we measure, and the practical difficulty it creates is not that marginal gains go negative. It is that they become *small* — near zero and comparable to the error of estimating them — while out-degree’s association with them reverses. That is why the sign flip is the phenomenon worth reporting and why $MC \geq 300$ is a precondition for reporting it at all.

Density is not the explanation. It is tempting to attribute the sign flip to density. We do not. Across the eight graphs the rank correlation between $\langle k \rangle$ and the saturated ρ_{degree} is -0.286 , which is weak, and the relationship is not monotone in density: NetHEPT (4.23) is positive, p2p-Gnutella08 (6.59) and Wiki-Vote (29.15) are only mildly negative, ca-GrQc (11.06) and Facebook (43.69) are strongly negative, and Congress-Twitter (55.95), the densest graph, is nearly zero at -0.047 . The defensible claim is about the *saturated regime*, not about dense graphs.

A structural hint. After in-weight normalisation, exposure δ is driven by a node’s in-degree while a seed’s value is driven by its out-degree. The two coincide only when a graph is approximately symmetric. Consistently, the graphs with $\rho(\text{in}, \text{out})$ near 1 show strongly negative out-degree correlations, while NetHEPT, whose in- and out-degrees are nearly independent, is the one graph where degree survives. We record this as a hypothesis rather than a result; testing it needs more graphs than the eight available here.

7 A Training-Free State-Conditioned Score

7.1 Definition

We use a state-conditioned closed form as a proposal signal, not as an unbiased estimator of the true marginal gain. Given the realised exposure state, let $g(e) = 2e(1-e)$ and let $N^+(w)$ be the out-neighbours of a candidate w , with edge weight ω_{wv} . Adding w raises the exposure of v by

$$\Delta e_w(v) = \omega_{wv}(1 - \delta(v)). \quad (6)$$

The one-hop proposal score is

$$\text{score}(w) = \sum_{v \in N^+(w)} [g(\delta(v) + \Delta e_w(v)) - g(\delta(v))]. \quad (7)$$

We also use a two-hop version, denoted δ_2 , in which the exposure increment is propagated once through the out-edges of each affected node, with the same $(1 - \delta(\cdot))$ attenuation.

The score has two useful properties. It is training-free and requires no candidate-level Monte-Carlo trials after the exposure state is available. It is also non-monotone: a candidate that pushes an already exposed neighbour past its window can receive a negative score. These properties make it a natural proposal signal for the corrected regime sweep in Section 6.

7.2 Scope of the score

The closed form is not asserted to be an unbiased estimator, a certificate, or a replacement for the true objective. Its current evidence is limited to the MC=300 rank comparison in Section 6; all earlier calibration and policy-comparison tables were produced before the state-machine correction or below the required Monte-Carlo budget and are omitted. Whether δ_2 can support a reproducible quality-cost advantage in sequential seed selection must be established by a new post-fix experiment with Full-MC, degree, random-pruning, repeated candidate pools, and explicit state-acquisition accounting.

8 Post-fix Experimental Status

The first implementation of this study produced several numerical tables before the activation state machine and the measurement contract had been audited. Those tables are deliberately not included in this submission: they used the pre-fix state machine, mixed incompatible cost units, or estimated saturated marginal gains with too few Monte-Carlo trials. Keeping invalid tables in the paper would make them easy to quote as results and would also obscure the corrected evidence.

The valid post-fix regime sweep is reported in Section 6. It uses paired Monte-Carlo trials at $MC = 300$, reports the seed fraction $|S|/n$, and records the per-candidate paired standard error. It establishes the regime-level sign pattern and corrects the earlier negative-share claim.

The sequential selection and sample-cost comparison is intentionally left as an open experiment. Before it can support a sample-efficiency claim, it must use the frozen contract, charge state acquisition, repeat candidate pools, include degree, δ_2 , Full-MC and random-pruning controls, and report paired quality and oracle cost. The corresponding artifact and protocol are listed in the project research state; no pre-fix baseline number is used here.

This separation is important: the analytic coverage counterexample and the corrected regime characterisation are current results, whereas a post-fix quality-cost result is not yet established.

9 Limitations and Threats to Validity

No claim of a learned advantage yet. The current evidence does not establish that a learned state-conditioned scorer improves on degree or the training-free δ_2 score. The earlier predictor and policy-comparison measurements were generated before the state-machine correction or under an insufficient Monte-Carlo budget, so they are not reported here. A valid learned-advantage claim requires held-out contexts, post-fix labels, repeated candidate pools, and downstream quality-cost evaluation.

The sign pattern is graph dependent. The corrected eight-graph sweep finds negative degree correlation in seven saturated graphs, with NetHEPT as an exception. We do not attribute this to density: the density-correlation association is weak and non-monotone. The plausible in-/out-degree explanation remains a hypothesis, not a theorem, and larger graph families are needed.

Scale and dataset coverage. The largest graph in the current sweep has 15,233 nodes. We do not demonstrate million-node behaviour, and the cost of obtaining the exposure state-one cascade in the current prototype-must be re-evaluated at that scale.

The surrogate is not a certified algorithmic route. The source model's $\lambda = \sigma^\kappa - \sigma^\tau$ is a difference of two monotone submodular functions. Each component can individually admit a coverage argument, but the difference need not be monotone or submodular. The pre-fix numerical surrogate-gap table is omitted; whether the source theorem is tight under its intended coupling remains open.

Eight protocol confounds. The audit found the following issues. They are listed because the protocol, not a pre-fix number, is the reproducibility contribution of this section.

1. **Stopping under a budget of at most k .** The stopping rule is now explicit and is recorded per decision; a policy that cannot evaluate a requested rule refuses it.
2. **State acquisition cost.** The exposure state is charged as part of the same cascade budget, rather than being silently treated as free.
3. **Noisy reference naming.** A Monte-Carlo greedy policy carries its MC budget in its name and is not called an exact oracle.
4. **Seed-size pooling.** State-dependence ratios are computed inside one seed size and reported per size; unsaturated contexts are not pooled with saturated ones.
5. **Candidate-pool selection.** Candidate pools are degree-stratified and repeated so that a single draw cannot determine the sign of a headroom claim.
6. **State deduplication.** Splits are guarded by the canonical seed-set/candidate-set state rather than relying on context identifiers alone.
7. **Label uncertainty.** Paired standard errors are recorded for marginal labels, and rank correlations in the saturated regime are not reported below MC=300.
8. **Weight normalisation.** The chosen strategy is recorded explicitly; sensitivity to `sum_to_one` versus `clip_to_one` remains required before claiming external validity.

The main limitation is therefore not hidden in an omitted table: the corrected regime result is valid, but a post-fix sequential sample-efficiency result has not yet been established. The paper’s title and algorithmic claims must be revisited if the pending quality–cost gate fails.

10 Conclusion

We set out to select seeds under a threshold-dependent overexposure model and found that the coverage representation behind the dominant algorithmic tool does not apply. Reverse-reachability sampling rests on the identity $F_D(S) = c \cdot \Pr[S \cap R \neq \emptyset]$ for a random set R independent of S , a function that is necessarily non-decreasing and submodular. On a one-hop instance of two candidates of weight 0.4 into one target, the model’s own window distribution gives $F_D(\{a\}) = 0.48$ against $F_D(\{a, b\}) = 0.32$, so the target is non-monotone and no such coverage function matches it. We are explicit about the limits of that statement: it rules out the standard coverage form and nothing more. It does not rule out signed decompositions, algorithms for restricted instances, or Monte-Carlo estimation, and it says nothing about whether learning helps. It also holds only for a named threshold distribution — under uniform-threshold linear thresholds the same instance is monotone. An earlier draft argued the stronger claim that "RR sets are empty"; that probe was mis-specified and the claim is withdrawn.

What replaces it is a re-measured regime characterisation and a measurement protocol. We identified the seed *fraction* $|S|/n$ as the variable that governs how a candidate’s value is ranked, and re-measured the sweep under the corrected

activation rule at 300 paired Monte-Carlo trials per cell: across eight graphs spanning mean degree 4.2 to 56.0, out-degree is negatively correlated with the true marginal gain on seven of them once $|S|/n \geq 20\%$, and a training-free state-conditioned closed form is the better ranker on all eight (§6). We also had to withdraw one column of that claim. An earlier version reported that 8–35% of candidates have a negative marginal gain; at 300 trials the mean is 0.5% and the worst cell is 4.0%. The sign flip is real and the negative-marginal share was Monte-Carlo artefact, and only adequate power separates the two.

The baseline comparison that an even earlier version built on top of the regime characterisation is omitted from this submission. It was produced by an activation rule that forbade an over-exposed node from turning negative and at Monte-Carlo budgets below the threshold the measurement itself requires. We draw the general lesson in §5 as five checks; the pre-fix numerical comparison is not evidence for the current paper, and re-evaluating a node’s state every round is the check that invalidated it.

The negative result that survives is about the reference itself. The upper bound the source model optimises is a *difference* of two monotone submodular functions, and submodularity is not preserved under differences, so even a valid upper bound would not by itself license a sampling-based guarantee. Whether that bound is tight anywhere, and whether the marginals it must resolve are in fact smaller than the estimation noise, are empirical questions we can no longer answer from the measurements we have.

Two directions follow directly. The first is to re-derive the sequential quality-cost comparison under the frozen model contract — a declared target set, seed eligibility, a budget of *at most* k , a priced state acquisition, and a Monte-Carlo reference with a stated tolerance — with the eight confounds of §9 removed. The second is to look for a graph-level criterion — we suspect in-/out-degree coupling — that predicts when degree-based heuristics fail, which would let a practitioner decide whether the overexposure regime needs to be handled at all.

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